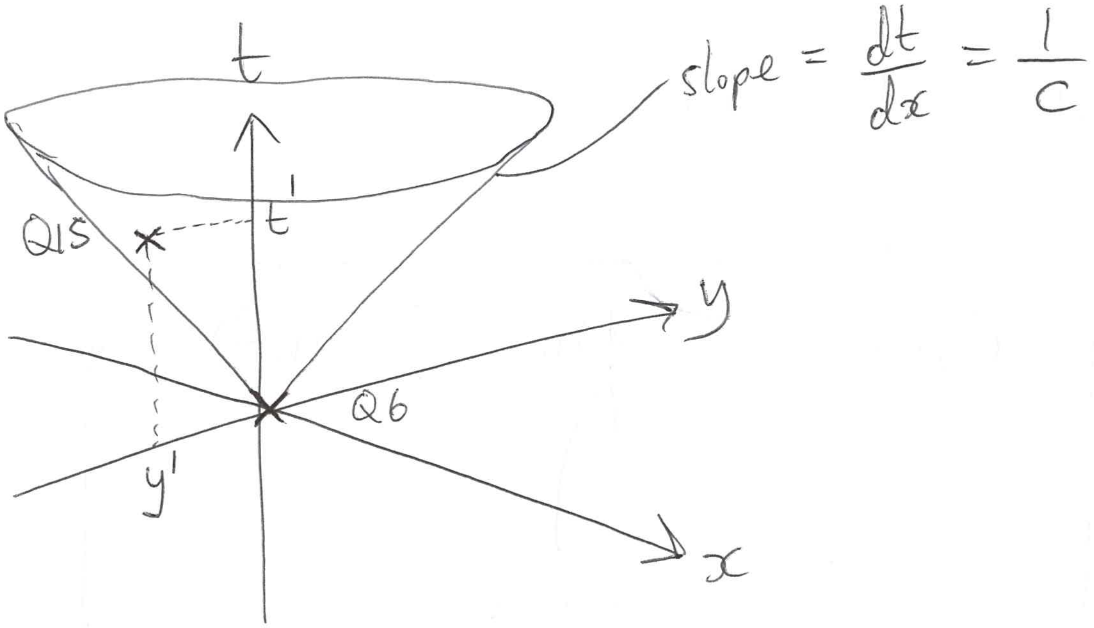


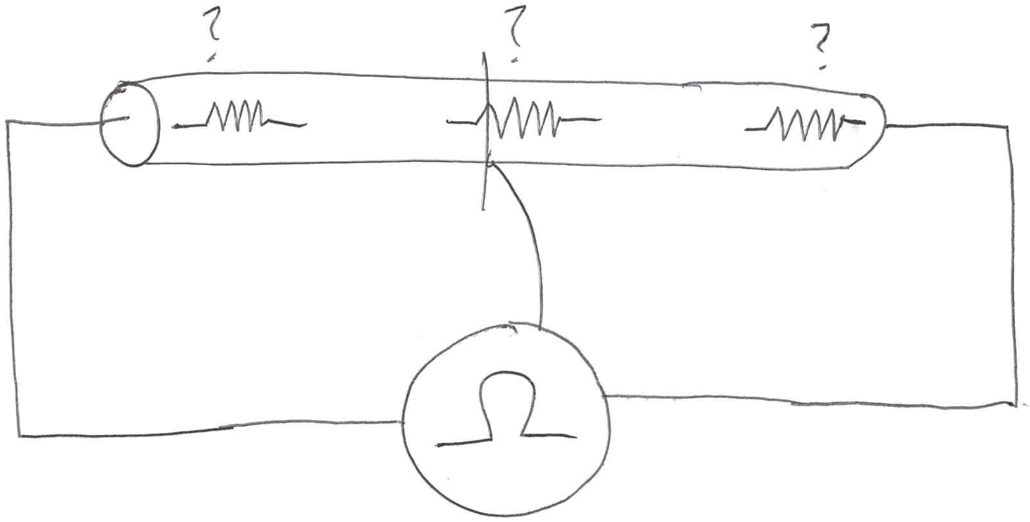
At $t=0$: Signal leaves Q6

At $t=t'$: Signal reaches Q15

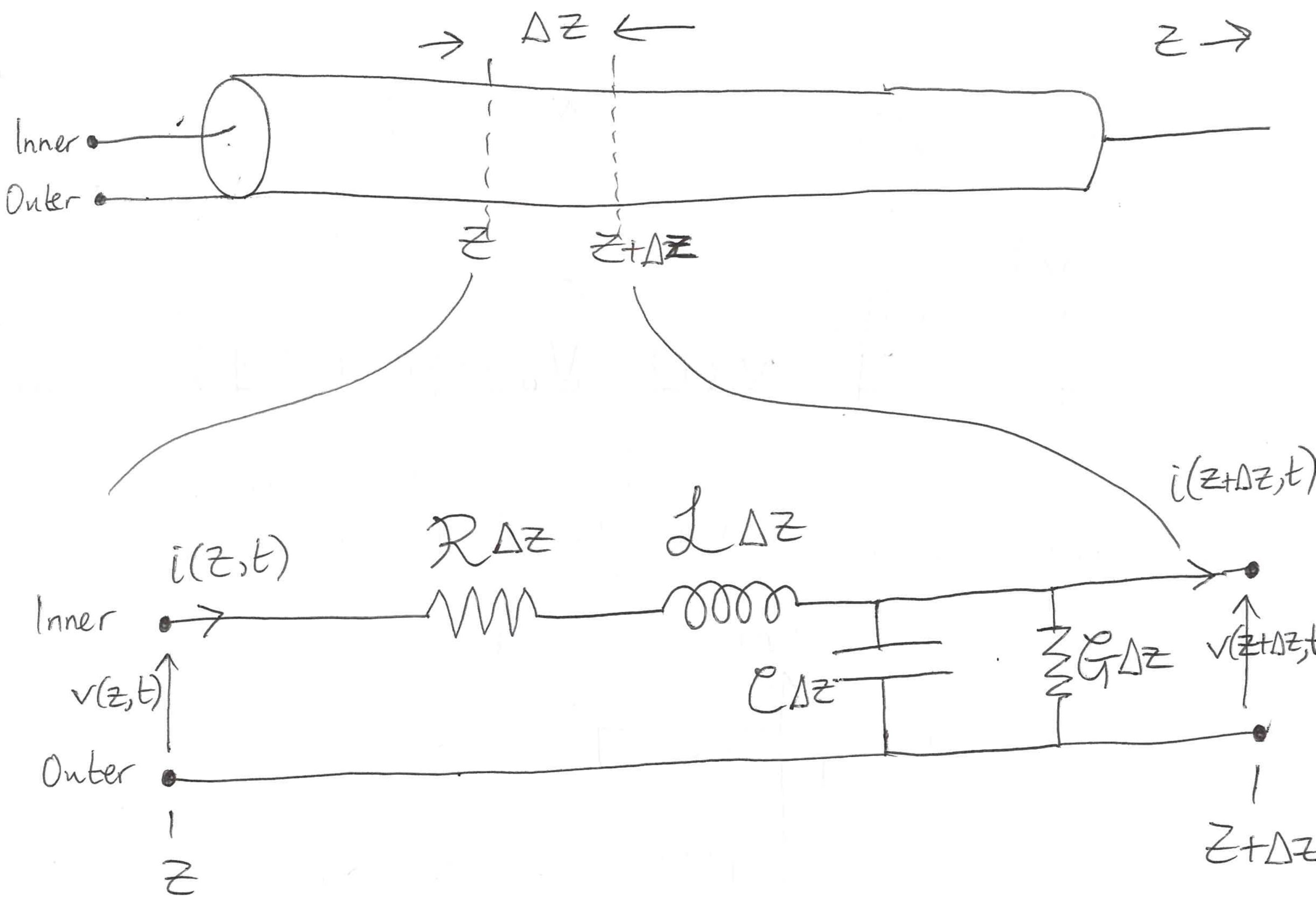
$$y' \leq ct'$$

$$L_e = \frac{L}{c\tau} \ll 1 \Rightarrow$$ lumped elements relevant

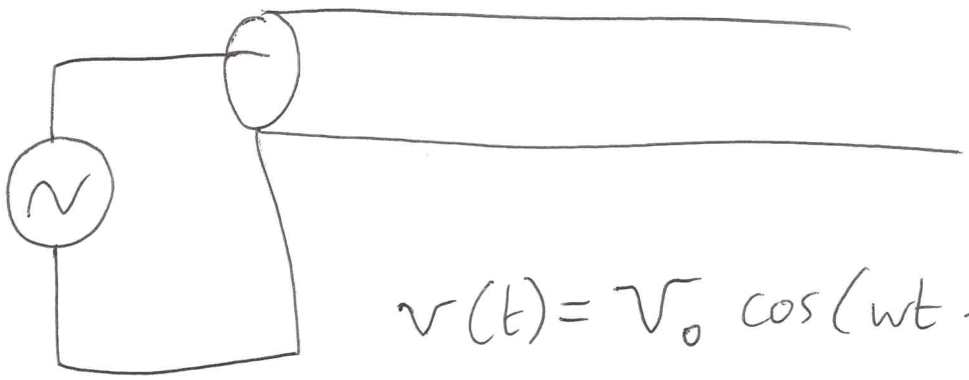
Distributed elements:



$$[R] = \Omega/m$$

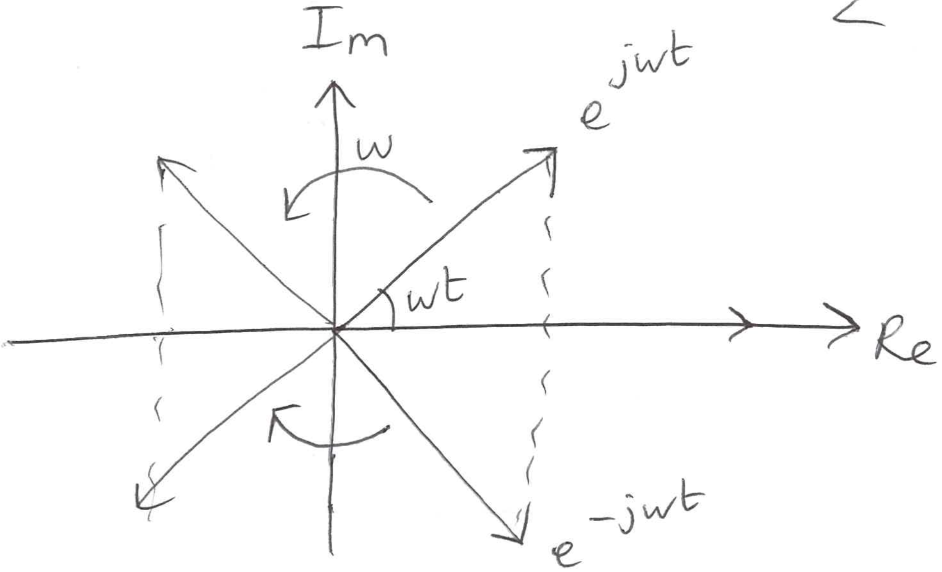


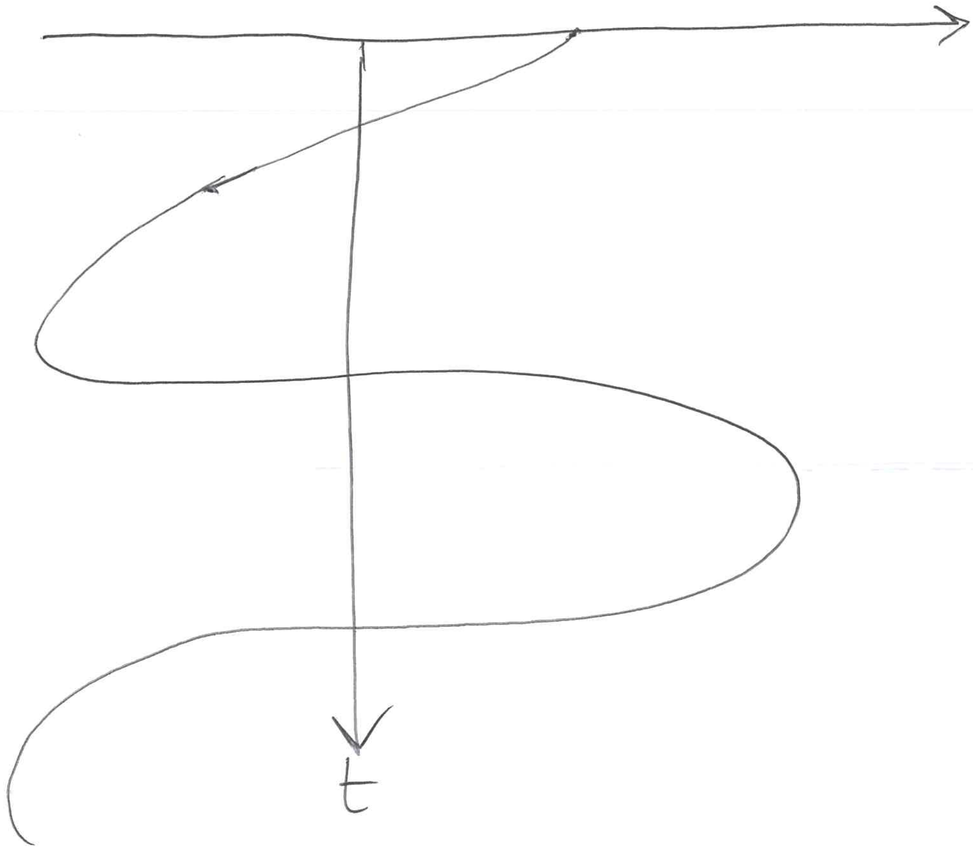
Perfect cable: $G = R = 0$



$$v(t) = V_0 \cos(\omega t + \phi)$$

Euler's formula: $\cos(\omega t) = \dfrac{e^{j\omega t} + e^{-j\omega t}}{2}$

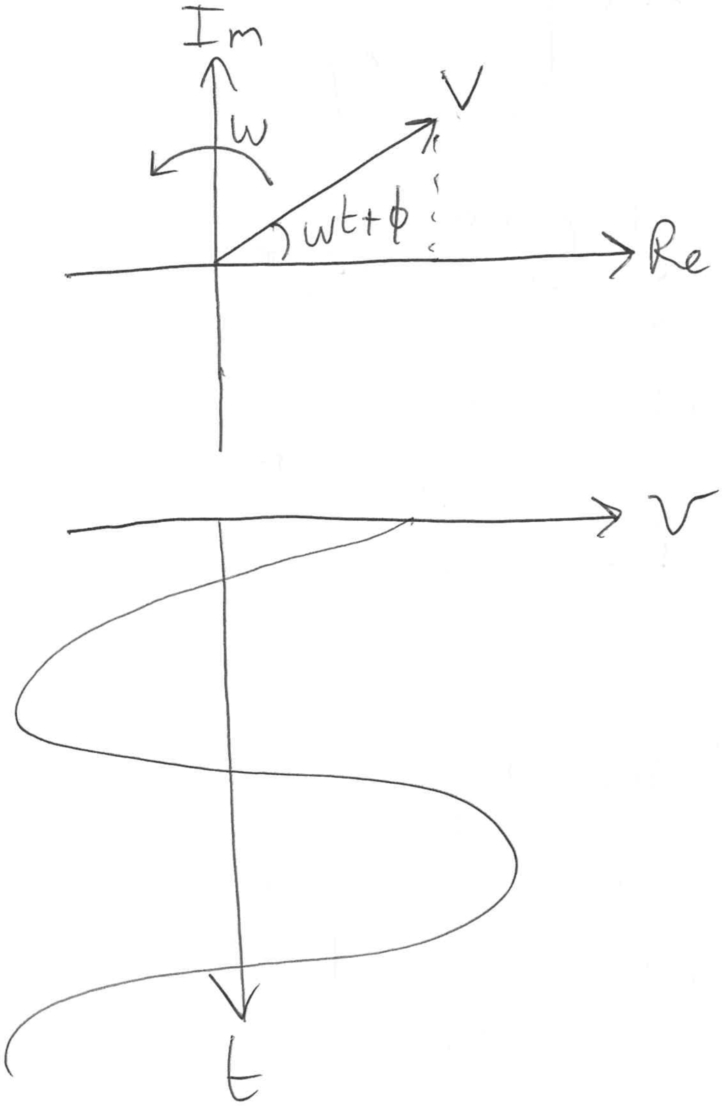




$$v(t) = V_0 \cos(\omega t + \phi)$$

$$e^{jx} = \cos x + j \sin x$$

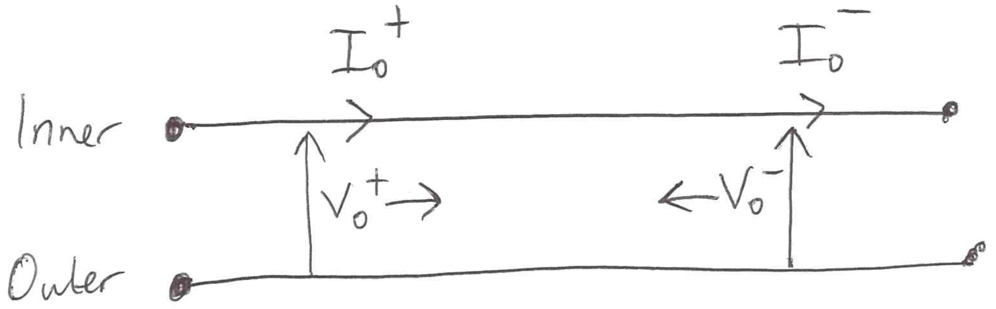
$$v(t) = \mathrm{Re}\left\{ V e^{j(\omega t + \phi)} \right\}$$

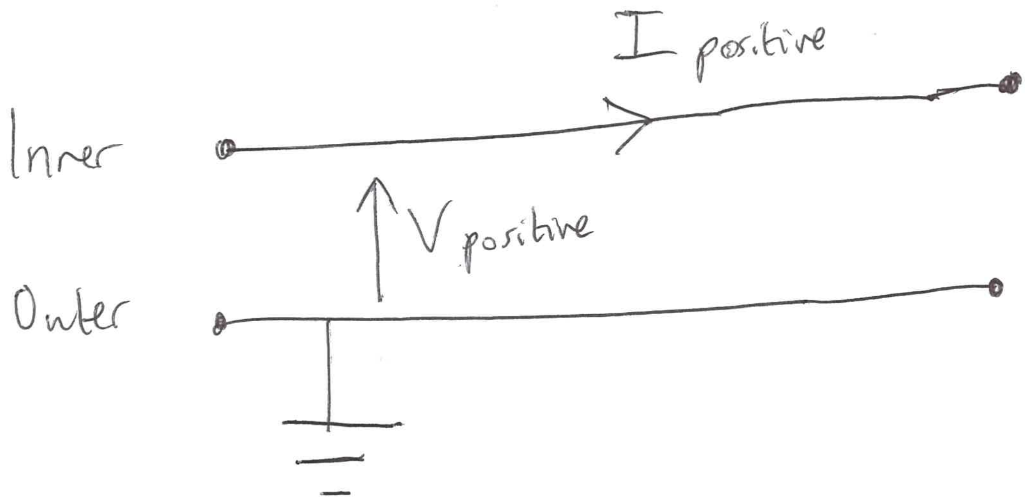


$$\frac{dv(t)}{dt} = \mathrm{Re}\left\{ \frac{d}{dt} V e^{j(\omega t + \phi)} \right\}$$

$$= j\omega \, \mathrm{Re}\left\{ V e^{j(\omega t + \phi)} \right\}$$

$$\therefore \frac{dV}{dt} = j\omega V$$





$$\frac{d^2V(z)}{dz^2} = \gamma^2 V(z)$$

$$V(z) = V_0^+ e^{-\gamma z} + V_0^- e^{+\gamma z}$$

$$\frac{dV(z)}{dz} = -V_0^+ \gamma e^{-\gamma z} + V_0^- \gamma e^{+\gamma z}$$

$$\frac{d^2V(z)}{dz^2} = \gamma^2 V_0^+ e^{-\gamma z} + \gamma^2 V_0^- e^{+\gamma z}$$
$$= \gamma^2 V(z)$$